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MILITARY SPECIFICATION

PROPELLING UNITS, OUTBOARD, DED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers deck mounted marine type diesel-engine-driven (DED) outboard propelling units, for propelling barges and pontoons.

1.2 Classification. Outboard propelling units shall be one of the following sizes, as specified (see 6.2):

Size 3,000 - 3,000 pound (lb) thrust.
Size 4,500 - 4,500 lb thrust.
Size 7,500 - 7,500 lb thrust.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

O-E-910 - Extinguisher Fire, Carbon Dioxide (Hand and Wheeled Types).
UU-T-81 - Tags, Shipping and Stock.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-P-40 - Packaging and Packing of Hand Tools.
PPP-T-60 - Tape: Packaging, Waterproof.

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MIL-C-104 - Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116 - Preservation-Packaging, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Flexible.
MIL-R-196 - Repair Parts for Internal Combustion Engines, Packaging of.
MIL-P-514 - Plates, Identification, Instruction and Marking, Blank.
MIL-T-704 - Treatment and Painting of Materiel.
MIL-L-2105 - Lubricating Oil, Gear, Multipurpose.
MIL-C-3774 - Crates, Wood; Open, 12,000- and 16,000-Pound Capacity.
MIL-E-10062 - Engines, Preparation for Shipment and Storage of.
MIL-E-11276 - Engines, Diesel: Industrial, Medium and High-Speed, General Specification for.
MIL-E-15130 - Enamel, Ship, Exterior, Alkyd, Haze Gray, No. 27, Formula No. 5H.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS (NBS)

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

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AMERICAN GEAR MANUFACTURERS ASSOCIATION (AGMA)

Gear Standards.

(Application for copies should be addressed to the American Gear Manufacturers Association, No. 1 Thomas Circle, N.W., Washington, DC 20005.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A36 - Specification for Structural Steel.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J516 - Standard for Hydraulic Hose Fittings.
J517 - Standard for Hydraulic Hose.
J534 - Standard for Hydraulic Lubrication Fittings.
J744 - Standard for Hydraulic Power Pumps.
J745 - Recommended Practice for Hydraulic Power Pump Test Code.
J755 - Standard for Marine Propeller-Shaft Ends and Hubs.
J931 - Information Report for Hydraulic Power Circuit Filtration.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., Two Pennsylvania Plaza, New York, NY 10001.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC. AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N. W., Washington DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

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3. REQUIREMENTS

3.1 Description. The propelling unit shall consist of a heat exchanger cooled marine diesel engine, marine reversing transmission, outboard drive assembly with propeller, and related equipment, propeller elevating mechanism(s), steel framework deck mounting assembly, and an integral control station with provisions for remote control.

3.2 First article. When specified (see 6.2), the contractor shall furnish a complete propelling unit for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Standard commercial product. The propelling unit shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product with any added features needed to comply with the requirements. Additional or better features which are not specifically prohibited by this specification, but which are a part of the manufacturer's standard commercial product shall be included in the propelling unit being furnished. Standard commercial product is a product which has been or will be sold on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model(s).

3.4 Identical items. All propelling units of the same size furnished under any specified contract shall be physically and mechanically identical. This requirement includes parts, assemblies, components, and accessories. No deviations will be acceptable without written approval of the contracting officer.

3.5 Material. Material shall be as specified herein. Material not definitely specified shall be of the quality regularly used by the manufacturer in the product he sells on the commercial market. Material shall be free from all defects and imperfections that might affect the serviceability and appearance of the finished product. Steel shall conform to ASTM Designation A36 for structural steel, and the related designations of table I therein, as applicable to other steel components.

3.6 Dissimilar metals. The use of dissimilar metals in intimate contact with each other shall be avoided to minimize galvanic corrosion. When the use of dissimilar metals cannot be avoided and injurious galvanic corrosion can be expected, an interposing material which will minimize or eliminate this effect shall be used.

3.7 Performance. The engine shall start and accept rated thrust in not more than 5 minutes at ambient temperatures from 0° Fahrenheit (F) to 110°F when operating in sea water. When the engine is operating, the overboard cooling water discharge shall flow in a continuous stream.

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When the engine is operated at the speed corresponding to the manufacturer's continuous duty rating, the bollard thrust shall be in accordance with table I. The engine shall automatically idle when the transmission is shifted to neutral between forward and reverse positions. The rotatable outboard drive assembly shall complete one 360° revolution in no longer than 6 seconds. When the propeller is raised above the waterline with full power throttle adjustment, the tachometer reading shall show not more than a 6 percent increase in speed. The propelling unit shall provide measureable thrust through all positions of the outboard drive unit from its normal operating position, to the point where the centerline of the propeller is at the waterline. Contact, by the propeller or any submerged part of the outboard drive assembly, with any solid submerged object shall cause the propeller leg assembly to be forced upward, on its radius, to a position that will clear the solid object. The outboard leg shall maintain the position to which it was forced until lowered by the operator. Forced cocking of the outboard leg shall not negate the steering and reversing capability of the unit when in its forced position. The size 7,500 unit shall provide not less than 80 percent of its rated thrust when the propeller is elevated from its normal operating position to the point where the bottom periphery of the propeller is even, or above a line extended from, and parallel to, the bottom of the hull of a pontoon barge assembly of P-series pontoons. A shear pin, slip clutch, or other suitable means shall be provided to prevent damage to the propeller drive mechanism in the event the propeller strikes an underwater obstacle with sufficient force to cause permanent damage to the drive train. The size 7,500 unit shall be capable of not less than 8 hours of continuous operation without damage or loss of function through a cycle of turns, reversing, beaching, debeaching, and crash stops.

Table I. Performance characteristics

Size (nominal)	Minimum acceptable thrust at rated engine speed (lb)
3,000	3,000
4,500	4,500
7,500	7,500

3.8 Design and construction. Propelling units shall be designed to provide long periods of reliable all-weather service. Unless otherwise specified (see 6.2), the maximum gross dry weight shall be 16,000 lb for size 3,000; 18,000 lb for size 4,500; and 20,000 lb for size 7,500. Unless otherwise specified (see 6.2), maximum overall dimensions, and elevation of propeller drive shaft for sizes 3,000 and 4,500 units shall be in accordance with figure 1. The width of the size 7,500 unit shall not exceed 6 feet 3 inches and the outboard leg length (depth), distance of outboard leg from the end of the barge, and overall length shall be as

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required, for the type and size of propeller utilized to meet the performance requirements of 3.7. All parts, components, and subassemblies, subject to service, repair, or replacement in the field shall be readily accessible, and conveniently secured with adequate protection from damage, parts and lubricant loss, and entrance of foreign liquid and material during continued rugged service. All screw threads shall conform to NBS Handbook H28. All moving and high temperature parts which may be hazardous to operating and maintenance personnel shall be enclosed or guarded. Each complete propelling unit shall be attached to the upper frame of a deck mounting assembly (see figure 2), with the lower frame bolted in place. When specified (see 6.2), additional lower frames shall be provided. The design and materials of the mounting frames and supports shall be of sufficient strength to withstand the dead weight load of the complete propelling unit, plus engine vibrational forces, plus a 1.5 shock loading factor, plus navigational and beaching forces, without damage or permanent distortion of the frame. Adequate wrench clearances shall be provided for removal and replacement of propelling units. Mechanical gearing and power trains shall be rated in accordance with AGMA Standards, at 150 percent of the input horsepower (hp) from the engine. Mechanical gearing and drives shall be fully enclosed in leakproof housings. Means shall be provided for filling, checking, and draining liquid lubricants and hydraulic fluid without disassembly. Outboard drive units, if disconnectable for shipping, shall be designed to prevent exposure of interior working parts and the loss of lubricants or hydraulic fluid. Hydraulic pumps, hydraulic motors, power hose, fittings, and controls shall conform to current SAE Standards, as applicable to the equipment and parts used. The essential subassemblies shall be engine with enclosure, heat exchanger, outboard drive, control station (port or starboard mounted), hydraulic starting system, hydraulic power distribution system, and related accessories. Engine, starter, heat exchanger, hydraulic reservoir, and related equipment shall be grouped with maximum practicable compactness on the deck mounting assembly framework, adjacent to the control panel, under a protective enclosure. The protective enclosure shall be designed to withstand the shocks of beaching and ocean wave splashing without engine failure or damage, and shall be removable for engine maintenance. Drain outlets for engine-heat exchanger coolant, crankcase oil, and hydraulic power distribution fluid shall be readily accessible for maintenance. Engine exhaust shall be discharged either upward at a point not less than 6 feet above deck level, or rearward at a point not less than 3 feet behind the control station.

3.8.1 Control station. Control station shall be complete with conveniently arranged controls, and instrument panel. When the controls are side mounted, the control station shall be designed so that it can be field converted from port to starboard or vice-versa. Each procurement will stipulate the control station location (see 6.2). Controls shall be provided for one man operation. Control arrangement and installation shall be designed for all weather day and night operation, and tactical

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blackout touch control. When specified (see 6.2), the control panel instrumentation shall be provided with a auxiliary marine type indirect lighting system, consisting of wiring, bayonent type bulbs and sockets, switch, and an independent 6- or 12-volt battery source within a water-tight enclosure. When an electric cranking and charging system is specified (see 6.2), the instrumentation lighting system shall be an integral part of the units electrical system. The transmission control shall be interlocked to prevent shifting into forward or reverse positions with engine speeds above engine idle speed. All instruments shall be weatherproof. All controls shall be identifiable by feel as to their function and position for blackout operation. Instruments and controls shall include not less than the following:

(a) Engine starting controls.
(b) Forward-neutral-rearward transmission control.
(c) Steering control.
(d) Outboard drive propeller elevation controls.
(e) Instrumentation as specified in 3.8.3.

3.8.1.1 Sound pressure levels. The sound pressure levels (SPLs), to which personnel may be exposed during normal operation of the unit, under all conditions specified herein, shall not exceed the values shown for internal SPLs in table II for the respective locations. Areas which personnel need not enter during normal operation, but in which the SPLs exceed the values shown, shall have a conspicuous warning sign as follows:

WARNING

EXCESSIVE SOUND PRESSURE LEVELS WHEN OPERATING

USE EAR PROTECTIVE GEAR

The propelling unit shall have sound attenuating accessories to reduce the external SPLs, at 100 feet radial distance from the set, to values which do not exceed the values shown for external SPLs in table II.

Table II. Maximum SPLs in decibels (db) (reference 0.0002 microbar)

	Octave band midfrequency (Hz)							
Receiver location	63	125	250	500	1000	2000	4000	8000
Internal:								
Operator's position	82	76	72	65	64	64	65	67
Other work areas	98	92	87	80	80	80	80	80
External: (100 ft radial distance)	77	67	58	55	50	47	45	40

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3.8.1.2 Fire extinguisher. When specified (see 6.2), a standard charge, squeeze grip type valve, 5-pound portable, carbon dioxide fire extinguisher conforming to O-E-910, type I shall be provided. The extinguisher shall be provided with a vehicle type mounting bracket mounted within reach of the operator when at the control station.

3.8.2 Outboard drive assembly. The outboard drive assembly shall consist of an enclosed vertical drive shaft or hydraulic power hose assembly, rotatable through 360°, with a 90° angle gear drive, or a hydraulic motor drive, and a propeller. A 360° rotation of the outboard drive shaft for steering shall be provided by hydraulic power transmission and motor. Hydraulic ram and piston control rods, when furnished, shall be made of suitable alloy, or heavily chromium plated, to preclude atmospheric corrosion or electrochemical decomposition. The assembly shall be propeller driven, either through drive shaft and bevel gearing or universal power transmission, or by directly connected hydraulic motor. Power transmission gearing shall be designed for use in either direction of rotation. Gear ratios or hydraulic motor speeds, shall be selected to obtain propeller rotative speeds as required to provide the thrust as applicable to the propelling unit size. The outboard drive assembly shall be provided with an oil lubrication system specifically designed for the application furnished. A convenient means of checking level of oil, draining, and filling shall be provided. Service shall not require disassembly. The outboard drive assembly shall be leakproof under all conditions of service. Disconnectable outboard drive assemblies shall be provided with oil seals and barriers to prevent loss of lubricant, or caps and plugs for hydraulic hose as applicable, when disassembled or shipped.

3.8.2.1 Obstacle clearing provisions. The size 7,500 propelling unit shall have two obstacle clearing capabilities. One shall be a planned action, utilizing the hull of the barge as a protector of the propeller, by elevating the propeller, prior to beaching or operating in shallow water (see 3.7). The second capability, which is required on all units, shall be an automatic action caused by contact with a solid submerged object, causing the outboard leg to swing upward on its radius to clear the object. The outboard leg shall maintain this forced position until lowered by the operator. A method for manual operation of the obstacle clearing devices shall be provided for use during engine shut down.

3.8.2.1.1 Outboard leg cockup. The outboard leg assembly shall have the ability to be cocked up, hydraulically or manually, to not less than 160° from its normal operating position (against the stops) to allow maintenance or repair function without docking.

3.8.2.2 Steering. The steerable vertical column shall be mounted in such a manner that the column and propeller assembly can be rotated a full 360° about the vertical axis. Full power of the propeller shall be

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available at any degree of steering rotation. Steering shall be accomplished by power actuated hydraulic motor with worm gear drive, and shall be controlled by handwheel at the control station. Auxiliary hand steering means shall be provided for emergency operation. An accurate mechanical navigational direction indicator, with the outline of a propeller embossed or indented for touch reading during blackout conditions, shall be linked to the steering mechanism and shall have its indicating dial mounted at the control station. In lieu of a direction indicating dial mechanism, the steering wheel may be interlocked with a one to one turning ratio, and provided with an index spoke or hand and finger knob, set in the UP position for forward travel and the DOWN position for rearward. Powered steering linkage shall be disconnectable for manual operation.

3.8.2.3 Propellers. Propellers shall be of manganese bronze, stainless steel, or other commercially acceptable material and shall be bored and fitted with a suitable gasket, packing, or rope guard at the propeller shaft seal. Size and number of blades shall be as required for the applicable thrust. When propeller shaft ends and propeller hubs are mated they shall conform to SAE J755.

3.8.3 Diesel engine. The engine shall be a standard commercial marine type diesel engine with a sea water heat exchanger cooling system. The engine furnished shall have the hp, torque, and speed characteristics at the manufacturer's continuous duty rating, to meet satisfactorily all the equipment performance requirements specified. The engine shall be furnished with the following accessories and equipment:

(a) Full flow oil filter.
(b) A fuel tank with capacity for not less than 8 hours of operation at maximum engine operating conditions.
(c) A T-valve in the fuel supply line with capped side inlet connection for reserve tank connection.
(d) Fuel priming pump.
(e) Starting and priming systems.
(f) Heat exchanger cooling system with the engine liquid coolant temperature thermostatically controlled within a 170° - 180°F range.
(g) Air cleaner located to prevent the entrance of water.
(h) A hydraulic cranking system with engine driven pump and emergency manual pump.
(i) High cooling liquid temperature warning device.
(j) Low lubricating oil pressure warning device.
(k) Manually operated, emergency overspeed shutoff control within convenient reach of the operator at the control station.
(l) An instrument panel complete with a lubricating oil pressure gage; cooling liquid temperature indicator; mechanical

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tachometer with continuous, and maximum engine speed rating conspicuously indicated; an engine hour meter; a mechanical type fuel gage mounted on the fuel tank, and a mechanical engine hour meter with 100-hour capacity and provision for resetting.

(m) A corrosion-resistant dry type muffler suitable for marine duty.

(n) Vertical exhaust outlets shall extend not less than 72 inches above the pontoon deck, and shall be provided with self-closing rain flaps.

(o) Hydraulic power distribution system fluid pressure gage with normal pressure range indexed.

(p) Outboard drive lubricating oil pressure gage with normal pressure range indexed.

(q) A propeller reversing machanism operated from the control station.

(r) When specified (see 6.2), electric cranking, charging, and integral instrumentation lighting systems with a 12-volt marine type battery enclosed within a watertight enclosure.

3.8.3.1 Diesel engine fuel. When specified (see 6.2), the diesel engine supplied shall be operable using fuel conforming to MIL-T-5624, grade JP-5 in addition to its total performance fuel which shall conform to VV-F-800, grade DF-2.

3.8.3.2 Engine for Army procurements. When specified (see 6.2), the size 4,500 unit shall be powered by a diesel engine conforming to MIL-E-11276, and as specified herein.

3.8.4 Sea water heat exchanger. The engine cooling system shall include an engine attached sea water to fresh water heat exchanger, designed to operate as an integral part of the engine. The efficiency of the system shall be such that the temperature of the engine liquid coolant will not exceed 200°F, or the engine manufacturer's recommended maximum temperature, whichever is less, when the engine is operating at maximum speed and rated hp on available sea or fresh cooling water. A self-priming circulating pump designed for operation with not less than a 10 foot suction lift, suction and discharge piping, engine manifold cooling or shielding, and a duplex strainer with transparent shock resistant bowl and a long handle 3-way valve to assure nonclog continuous operation, shall be provided. The strainers shall be located at or near the control station to permit cleaning while underway. The cooling water discharge shall be above the water level where the stream can be seen by the operator. If required, hydraulic fluid cooling shall be provided.

3.8.5 Hydraulic systems. A complete hydraulic power distribution system shall be provided for obstacle clearing operation of the outboard

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drive and for steering control. The hydraulic power system shall consist of a hydraulic fluid reservoir, an engine driven hydraulic pump, filters, hydraulic motors, pressure relief and regulating valves, valve control levers, pressure gages, suction and pressure lines, piping, and emergency bypass valving where required. Brackets for rigidly mounting the various parts shall be furnished. Means for filling and draining, and checking fluid level in the reservoir shall be provided. Hydraulic fittings and hose shall conform to SAE J516 and J517 for use with a working pressure of 2,000 pounds per square inch (psi) on the hydraulic power system and 3,000 psi on the hydraulic cranking system. An independent hydraulic engine cranking system shall be provided. The system shall be designed to accomplish a cold weather start with not less than 1,500 psi, and to store not less than 3,000 psi with full accumulator capacity. Power pumps and hydraulic motors shall conform to SAE J744, and meet the designed pressure-flow requirements when tested in accordance with SAE J745. A hydraulic power circuit filtration system conforming to SAE J931 shall be installed. A method for manually charging the hydraulic cranking system shall be provided.

3.8.6 Electrical systems. When specified (see 6.2), complete marine type electrical systems shall be provided for cranking, charging, and instrumentation lighting of the propelling unit. The systems shall incorporate a 12-volt battery for electrical storage enclosed within a watertight enclosure. In addition, all required engine indicating and warning devices listed in subparagraphs (i), (j), and (l) of 3.8.3 shall incorporate electrical components wherever applicable in accordance with good commercial practice. All electrical components shall be of the type and quality utilized in good commercial marine applications for equipment of this type.

3.9 Lubrication. Lubrication means shall be provided for all moving parts requiring lubrication. Hydraulic lubrication fittings shall be in accordance with SAE J534. When normal lubrication pressures may damage grease seals and other parts, fittings of the pressure relief type shall be used. Force feed lubricating pressures shall be equally effective in forward and reverse gear operation. The propelling unit shall be lubricated prior to delivery with lubricants designated for use in the 0°F to 32°F temperature range. The propelling unit shall be conspicuously tagged to identify the lubricants and their temperature range.

3.10 Lifting and tiedown attachments. The propelling units shall be equipped with lifting and tiedown attachments. Lifting and tiedown attachments shall conform to MIL-STD-209, type I or type II. Attachments shall be indicated by a transportation plate conforming to MIL-P-514, and identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for tie down of the equipment on the carrier when shipped.

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3.11 Treatment and painting. All exterior surfaces shall be cleaned, treated, prime coated, and intermediate coated in accordance with MIL-T-704, type B. All such coated surfaces shall be finished with two coats of haze gray enamel conforming to MIL-E-15130 applied to a total overall dry film thickness of not less than 5 mils. All paint shall be applied to dry surfaces in a dry atmosphere. If spray or roller painting is utilized, all welds and corners shall first be thoroughly brush coated. The finished paint application shall be smooth in appearance, continuous in color, free of dry overspray, pinholes, orange peel, sags, or other film defects.

3.12 Instruction and warning plates. The propelling unit shall be equipped with instruction plates suitably located, describing any special or important procedure to be followed in operating and servicing the equipment. Plates shall conform to MIL-P-514, composition C, and shall be attached to the unit. Plates shall be of a material which will last and remain legible for the life of the equipment, and shall be securely affixed thereto with brass screws or bolts of not less than 1/8-inch diameter. When affixed to surfaces subject to painting, the plates shall be embossed for legibility after painting.

3.13 Identification marking. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.14 Stenciling. The gross weight of each propelling unit shall be stenciled on each side of the unit. Stenciling shall conform to MIL-T-704.

3.15 Workmanship. Defective components shall not be furnished. Parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished without the written approval of the contracting officer. Welded, bolted, and riveted construction utilized shall be in accordance with the highest standards of the industry.

3.15.1 Fabricated parts and components. Fabricated parts and components shall be free from discontinuities, cracks, welding undercuts, weld spatter, and any other injurious defects. Start and finish of hand and automatic welds shall be done in a manner to provide uniform welds. End weld metal in the high-stressed members shall be feathered into the parent metal of the parts being connected.

3.15.2 Subassemblies. Subassemblies shall be within manufacturing tolerances to permit proper fitting during assembly. Shims, spacers, calking compounds, and weld buildup shall not be used to make allowances for poor fitting assemblies.

3.15.3 Bolted and riveted connections. All bolt and rivet holes shall be accurately punched or drilled and shall have the burrs removed.

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Mating surfaces shall be flat and parallel and matching holes shall be concentric. Rivets shall be driven to completely fill the holes and each type head of the same diameter rivets shall be uniform in size and shape. All bolts, nuts, and screws shall be tight. Washers or lockwashers shall be provided.

3.15.4 Assembled propelling unit. The assembled propelling unit shall be free from any defects that could cause malfunctioning of components or interfere with safe operation. Construction and applied coating shall be compatible with the best quality workmanship for this type of equipment.

3.15.5 Welding. Surfaces to be welded shall be free from foreign matter which would be injurious to the weld. All welding procedures shall be in accordance with a nationally recognized code. Welds shall transmit imposed stresses without permanent deformation or failure when subjected to proof or service loadings.

3.15.6 Castings. All castings shall be sound and free from patching, misplaced coring, warping, and any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Examination. Each propelling unit shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the supplier's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.2.1 First article inspection. First article inspection shall be performed on one propelling unit when a first article sample is required (see 3.2). This inspection shall include the examination of 4.2 and the

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tests of 4.3. The first article may be a first production item or a standard production item from the supplier's current inventory provided the propelling unit meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining propelling units to be furnished under the contract.

4.3 Tests. Permanent records, available to the Government, shall be kept of all test results and performance measurements. Production units shall be tested in accordance with 4.3.1. The first article, when required (see 3.2), shall be tested in accordance with 4.3.2. All defects shall be corrected. Noncorrectable defects shall be cause for rejection under the terms of the contract. After the completion of each test, the oil in the outboard drive of mechanical transmissions shall be drained and examined for the presence of water. Presence of water constitutes failure of the test. Lubricating oil pumps and lubricating system in the mechanical outboard drives shall be bench tested during the initial assembly of all units. Loss of fluid in hydraulic gear motor drives shall be cause for rejection.

4.3.1 Production unit tests. Each propelling unit shall be tested for fit and function of all moving parts and subassemblies. Not less than one propelling unit of each size specified under each contract shall be tested for bollard thrust under actual or simulated conditions of operation.

4.3.2 First article tests. The first article shall be mounted on a pontoon barge assembly of not less than 21 (7 long by 3 wide) 5- by 5- by 7-foot pontoons (see figure 3), and loaded by cargo or flooding to obtain a freeboard between 18 and 24 inches (from mounting deck to water surface). The propelling unit shall then be operated under field conditions with not less than two per hour each of attempted beachings, tight turns, crash stops, and underway transportation. Unless otherwise specified (see 6.2), first article testing shall be continued for not less than 40 hours of 8 hours (average) per day performance. Each day of scheduled operations shall include not less than the following:

(a) One hour underway at the engine's rated speed.
(b) One-half hour at maximum speed.
(c) Four crash stops from maximum speed, accomplished by rotating the outboard drive 180° with the steering controls.
(d) Four crash stops from maximum speed, accomplished by shifting to reverse gear.
(e) Four beaching operations consisting of driving the pontoon assembly at maximum speed onto a sandy beach, followed by backing away into deep water. Beaching operations shall not be required for size 3,000.

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(f) Check operating time on the engine hour meter and cruise at random until the equivalent of 8 engine hours at rated engine speed have been completed each day, for a total of not less than 40 hours of operation without breakdown.

Observations shall be made to determine prompt starting, steady running, response to controls, and the proper functioning of every item specified herein. Any malfunction shall constitute failure of the test. The propelling unit shall be attached at a point in the plane of the centerline of the vertical outboard drive housing, to a stationary object, with a horizontal line of sufficient length to dissipate any backwash effect and incorporating a calibrated dynamometer. The outboard propelling unit shall be operated in forward gear at varying engine speeds up to full throttle. Engine speed versus bollard thrust (static pull) shall be recorded. Bollard thrust at the manufacturer's recommended engine speed for continuous duty shall be checked for conformance to the requirements as applicable to the size (see 1.2). Failure to attain this value shall constitute failure of the test.

4.4 Preparation for delivery inspection. The preservation, packaging, packing, and marking of the propelling units shall be inspected to verify conformance to the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly and matchmarking. Disassembly shall be the minimum necessary to safeguard parts known to be subject to damage or loss and to accomplish reduction in cube. Unless otherwise specified (see 6.2), the swing joint assembly and outboard drive shall not be removed. When specified (see 6.2), the swing joint assembly and outboard drive shall be removed by disconnecting associated couplings and removing the complete outboard drive from the bed of the power plant. Bolts, nuts, screws, pins, and washers removed shall be reinstalled in one of the mating parts and secured to prevent their loss. When necessary to facilitate reassembly, parts removed and mating parts shall be matchmarked. Matchmarking shall be printed on tags conforming to UU-T-81, type A.

5.1.1.2 Preservatives. Preservatives shall conform to the applicable specifications listed in, and shall be applied in accordance with MIL-P-116.

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5.1.1.3 Unpainted surfaces. Unpainted exterior ferrous metal surfaces including threaded surfaces and surfaces exposed by disassembly shall be coated with type P-1 preservative.

5.1.1.4 Machined surfaces. Exterior ferrous metal machined or critical surfaces of component parts and assemblies, such as shafts and gears exposed by disassembly shall be coated with type P-6 preservative. The preserved coated surfaces shall be wrapped or covered with barrier material conforming to MIL-B-121, type I, grade A, class 2, secured in place with tape conforming to PPP-T-60, type IV.

5.1.1.5 Engines. Engines, component parts, and accessories shall be preserved in accordance with MIL-E-10062, level A, type III.

5.1.1.6 Transmissions. The transmissions shall be filled to the operating level with lubricant conforming to the applicable lubrication order. Operate the transmission to assure coating of all interior surfaces with the lubricant. A tag conforming to UU-T-81, type A, shall be attached in a conspicuous location indicating that: THE HOUSING IS FILLED WITH THE REQUIRED LUBRICANT.

5.1.1.7 Enclosed gears. Gear housings containing lubricant conforming to the applicable lubrication order, or to MIL-L-2105 shall be checked for proper operating level and the gears actuated to insure coating of all interior parts and surfaces. Gear housings not containing these lubricants shall be drained, and refilled to the operating level with lubricant conforming to the applicable lubrication order and actuated as stated above. A tag conforming to UU-T-81, type A, shall be attached in a conspicuous location indicating that: THE HOUSING IS FILLED WITH THE REQUIRED LUBRICANT.

5.1.1.8 Exposed drive chains. Exposed chains shall be coated with enough type P-3 or type P-9 preservative to assure penetration of the preservative to the inner surfaces of the rollers, pins, and bushings. After the excess preservative has drained, the chains and sprockets shall be coated with type P-1 preservative.

5.1.1.9 Hydraulic systems. The hydraulic fluid supply tank shall be filled to the operating level with approved hydraulic fluid. Where practicable, pistons shall be in their fully retracted position, with any remaining unpainted surfaces coated with type P-1 preservative. Where pistons cannot be fully retracted, exposed areas of normally retractable surfaces shall be coated with type P-6 preservative. The type P-6 preservative coated surfaces shall be spirally wrapped with barrier material conforming to MIL-B-121, type I, grade A, class 2. The wraps shall be secured in place with tape conforming to PPP-T-60, type IV. All other

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unpainted exterior surfaces shall be coated with type P-1 preservative. All assemblies actuated by the hydraulic system shall be secured to prevent movement.

5.1.1.10 Fire extinguishers. The fire extinguishers shall be placed in the mounting brackets and secured.

5.1.1.11 Repair parts. Unless otherwise specified (see 6.2), repair parts shall be preserved and packaged in accordance with MIL-P-116. When specified (see 6.2), repair parts shall be preserved and packaged in accordance with MIL-R-196, level A; or where parts are not specifically covered in MIL-R-196, an applicable submethod of preservation in MIL-P-116 shall be used.

5.1.1.12 Maintenance tools. Maintenance tools shall be preserved and packaged in accordance with PPP-P-40, level A.

5.1.1.13 Technical publications. Technical publications for each propelling unit shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.1.1.14 Consolidated packaging. Small disassembled component parts and the repair parts, maintenance tools, and technical publications shall be packaged in close-fitting boxes conforming to PPP-B-621, class 1; or PPP-B-601, domestic type. The contents shall be cushioned, blocked, and braced to prevent movement within the boxes. Strapping shall not be required.

5.1.2 Level C The complete propelling units shall be preserved and packaged in accordance with the supplier's standard practice.

5.2 Packing. The packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The complete propelling units shall be packed in crates conforming to MIL-C-104, type II, style a, class optional. The contents shall be anchored, blocked, and braced in accordance with the crate specification.

5.2.2 Level B. The complete propelling units shall be packed in crates conforming to MIL-C-3774, type I, style A, or type II. The contents shall be anchored, blocked, braced, and waterproofed with a shroud in accordance with the crate specification.

5.2.3 Level C. The complete propelling units shall be packed in a manner which will insure arrival at destination in satisfactory condition

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and which will be acceptable to the carrier at lowest rates. Packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. Interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Propelling units are used on barges and floating pontoon assemblies for water transportation during amphibious landings to move and place heavy floating structures, such as pontoon causeways, and in general combat support operations.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of propelling unit required (see 1.2).
(c) When first article is required for inspection and approval (see 3.2, 4.2.1, and 6.4).
(d) Maximum weight and maximum overall dimensions other than as specified (see 3.8).
(e) When additional lower mounting frames are required (see 3.8).
(f) Control station location (see 3.8.1).
(g) When auxiliary control station panel lighting is required (see 3.8.1).
(h) When fire extinguisher is required (see 3.8.1.2).
(i) When complete cranking, charging, integral instrumentation lighting, indicating, and warning electrical systems are required (see 3.8.3(r) and 3.8.6).
(j) When engine conforming to MIL-E-11276 is required for size 4,500 (see 3.8.3.2).
(k) When other than 40-hour first article test is required (see 4.3.2).
(l) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).
(m) When the swing joint assembly and outboard drive are to be removed (see 5.1.1.1).
(n) When repair parts are to be preserved and packaged in accordance with MIL-R-196 (see 5.1.1.12).

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6.3 Cross reference of classifications. The following propelling units are functionally interchangeable with units shown from MIL-P-15916D and MIL-P-82022(YD).

MIL-P-15916E	MIL-P-15916D	MIL-P-82022(YD)
Size 3,000	Size 110	N/A
Size 4,500	Size 165	MIL-P-82022(YD)
Size 7,500	N/A	N/A
	Size 225 deleted	

6.4 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the Armed Services Procurement Regulations. The first article should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test, and approval of the first article.

6.5 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Armed Services Procurement Regulation, the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.6 Revision changes. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to extensive changes involved.

Custodians:

Army - ME
Navy - YD
Air Force - 82

Preparing activity:

Navy - YD

Project No. 2010-0004

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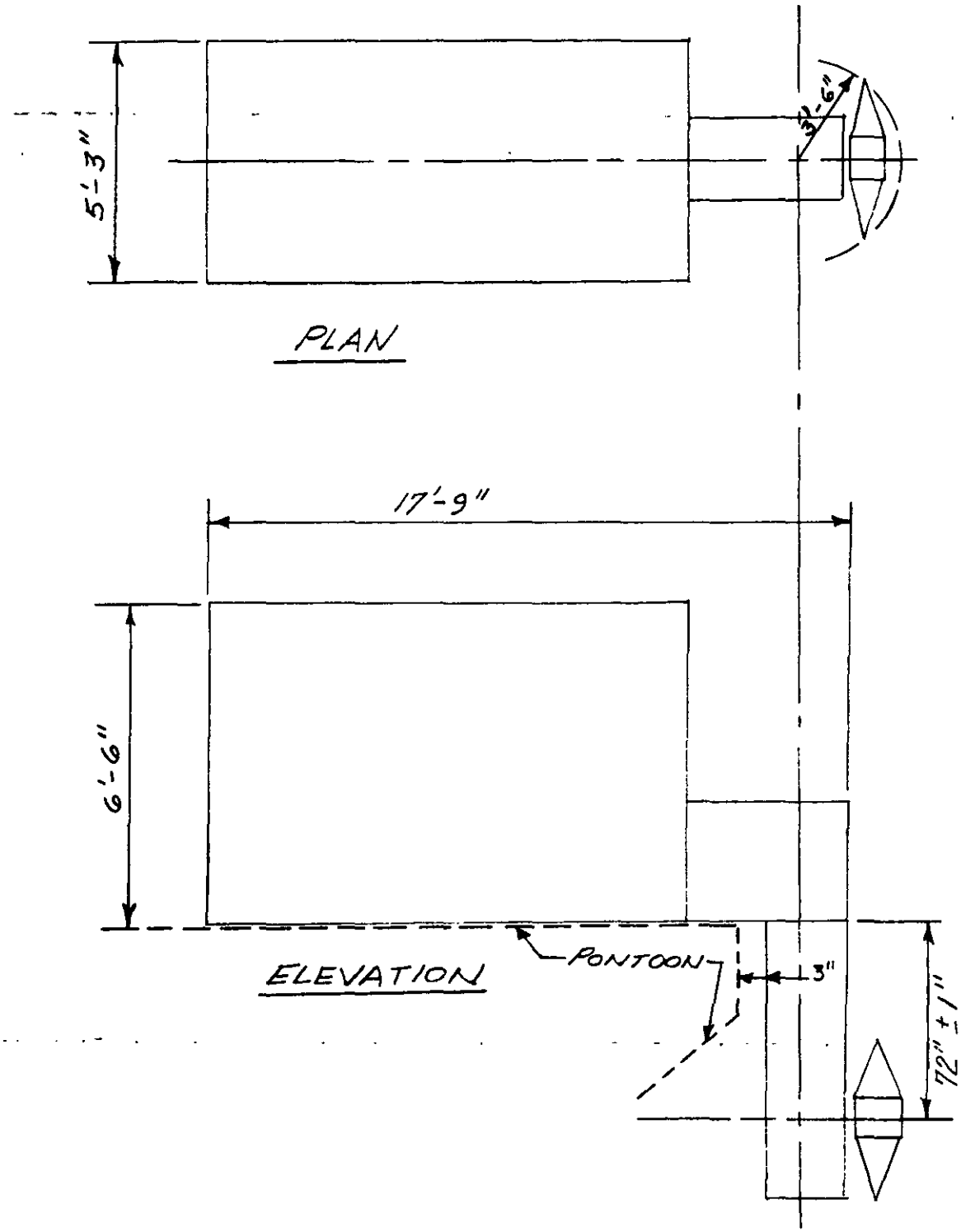


Figure 1. Maximum envelope dimensions, sizes 3000 and 4500

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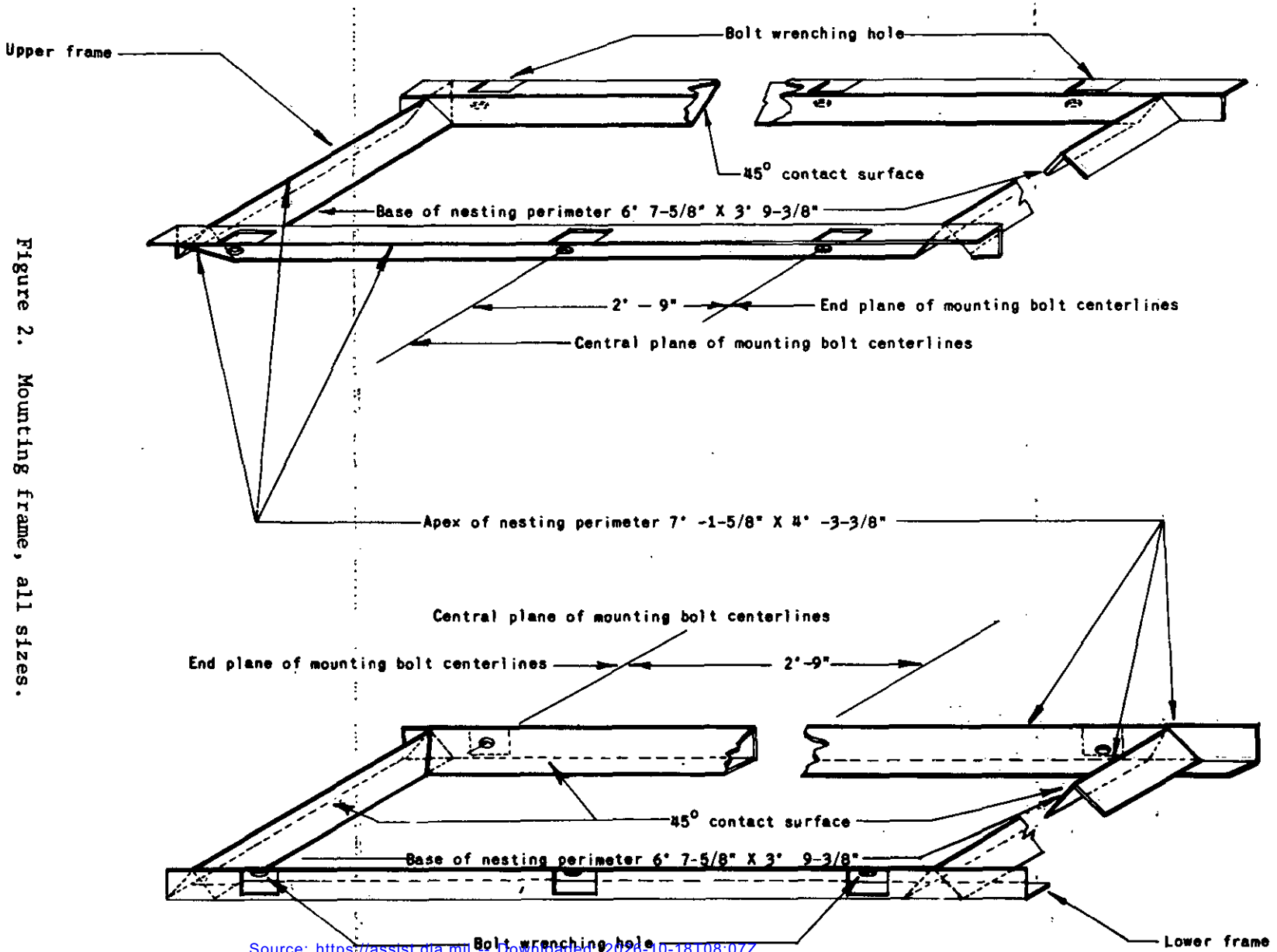


Figure 2. Mounting frame, all sizes.

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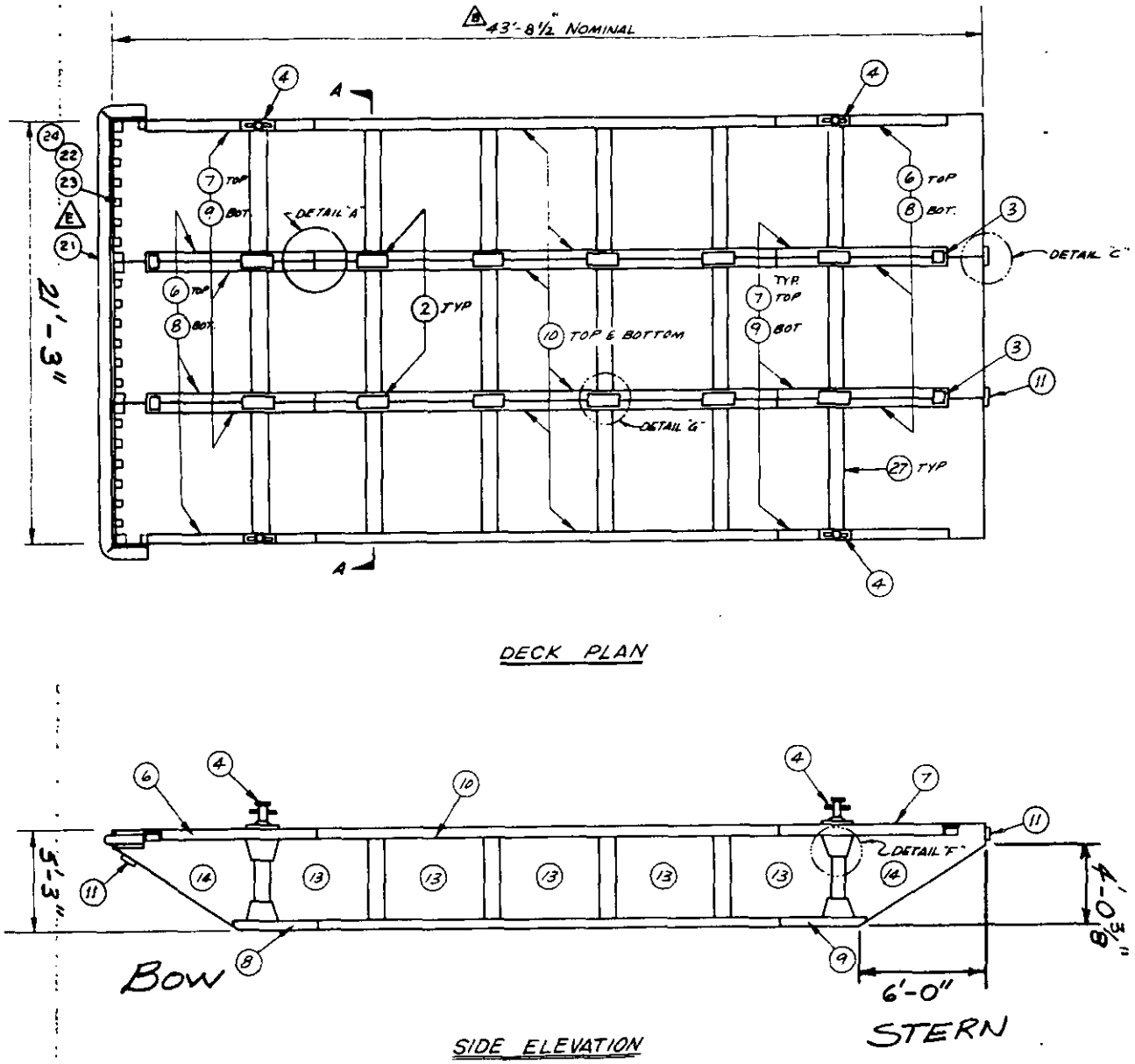


Figure 3. 3 by 7 Pontoon barge.

GPO 1974-603-125/767

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

2010-0004

DOCUMENT IDENTIFIER AND TITLE

MIL-P-15916E - PROPELLING UNITS, OUTBOARD, DED

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED S/N 0102-014-1802

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